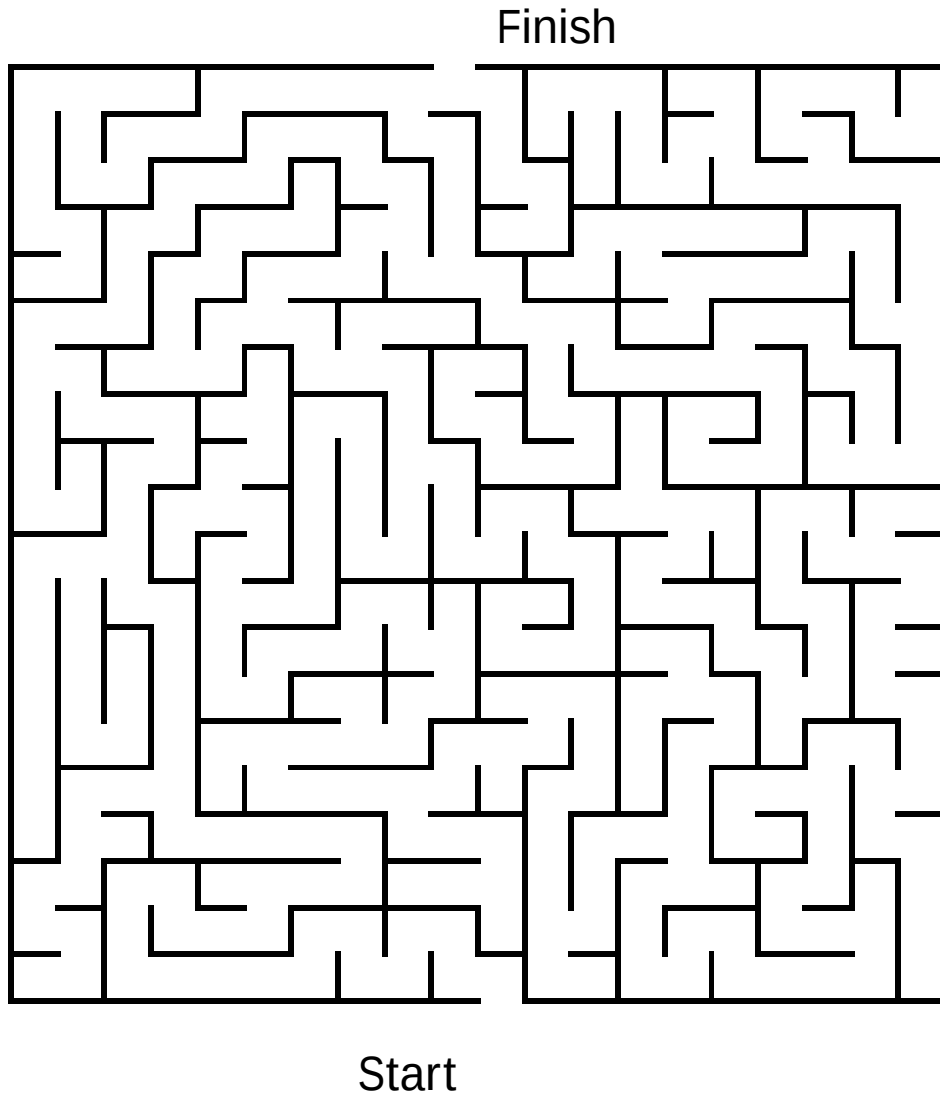




Lesson 04

Algorithms and Pseudocode

Create Algorithm to solve this maze



What might the Algorithm to solve this maze look like

- List step by step how to solve it.
- OR
- Create a set of rules for a “bot” to follow
- Some Advantages and Disadvantages of each.



Lesson Objectives

1. Understand what an algorithm is and its application in programming.
2. Learn and understand the concept of pseudocode.
3. Develop the ability to write simple algorithms.
4. Learn how to translate algorithms into pseudocode.
5. Understand the importance of efficient algorithm design.



What is an Algorithm?

- ♦ An algorithm, in relation to computers, is a defined set of step-by-step procedures or instructions that tells a computer exactly what to do to solve a specific problem or complete a particular task.
- ♦ List the steps needed to create a breakfast of Buttered Toast with Strawberry Jam
- ♦ Give some examples of how Algorithms are used in Computing



What is Pseudocode?

- ◆ Pseudocode is a simple way of writing programming code in English or in a way that can be understood by humans. It's not an actual programming language
- ◆ Example
 1. Start
 2. Type in 3 numbers
 3. Add the 3 numbers together
 4. $\text{average} = \text{sum} / 3$
 5. Print average
 6. End

Details of an Algorithm



◆Steps to Create an Algorithm

1. Clearly Define the Problem
2. Identify the Procedures
3. Order procedures
4. Check the Procedures
5. Test the Algorithm



Features of a Good Algorithm

1. Unambiguous
2. Well-defined Inputs and Outputs
3. Finiteness
4. Feasibility
5. Language Independent
6. Independent of Machine
7. Efficient

Purpose and properties of Pseudocode



1. **Clarity**
2. **Planning**
3. **Communication**
4. **Debugging**
5. **Simple**
6. **Language Independent**
7. **Ambiguity Free**
8. **High Level**

Rules of Pseudocode



These are more guidelines

1. Write one statement per line
2. Use Simple English
3. Indent for logic:
4. Keep it short
5. Be specific



Creating an Algorithm

START

1. Fill the kettle with water
2. Switch on the kettle
3. While waiting for the water to boil,
 1. Place a tea bag in the cup
 2. Add sugar to the cup (if needed)
4. When the kettle switches off, pour the hot water into the cup
5. Stir the tea
6. Wait for 2 minutes
7. Remove the tea bag from the cup
8. If preferred, add milk and stir the tea
9. The tea is ready for drinking

STOP

Convert to Pseudocode



```
◆ START
  Fill up kettle with water
  Turn ON Kettle
  Place teabag in cup
  IF tea requires sugar THEN
    ◆ Add sugar in the cup
  ENDIF
  WAIT until kettle switches OFF
  Pour water in cup
  Stir the mixture
  Wait 2 minutes
  Remove teabag
  IF milk needed THEN
    ◆ Add milk to cup
    ◆ Stir the mixture
  ENDIF
  PRINT "Tea is ready!"
◆ STOP
```